

Parameter Sweep Report: α , $\beta_{\min}$, c

Model: Reconstructed Ma et al. (2024) model · one sub-lethal dose ($a_0 = 2.15$, inhibitor $i = 10$) · $21 \times 21 \times 21$ grid = 9,261 simulations

Base parameters & variation range

Decoupling of the parameters

Sweep setup:

- α = growth of R relative to S (0.75–1); $\beta_{\min}$ = R's private protection (0 = strongest, 1 = none); c = inhibitor's intracellular efficacy (0–1).
- All other parameters are held fixed at the paper's Supplementary Information base values, we start with $S = R = 0.2$, nutrient = 4.
- Lysed cells return only 80% of their nutrient ($\xi = 0.8$), so total density at steady state (SS) is 3.0–4.3. The endpoints describe how this total is split between S and R.

1. Endpoint Analysis

1.1 Diversity Analysis

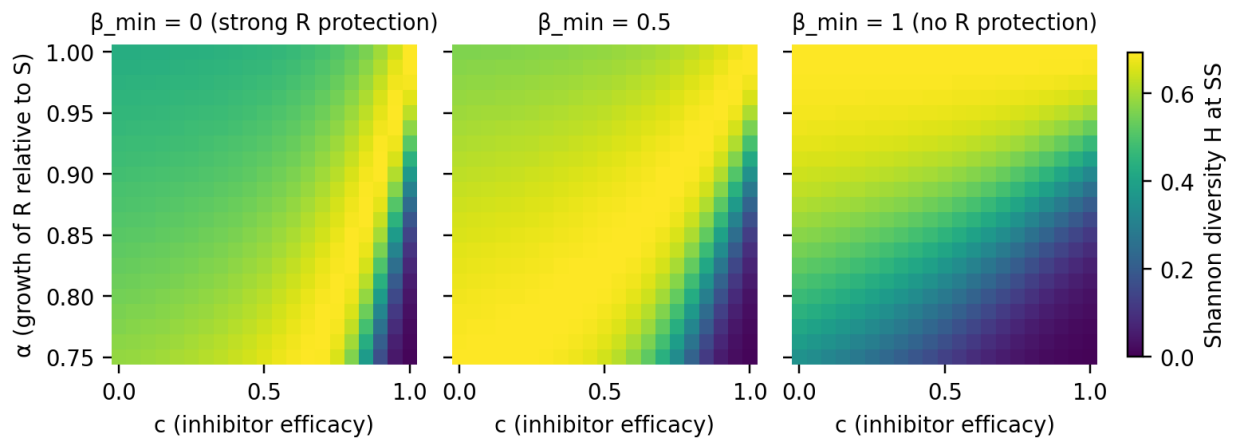


Figure 1. Shannon diversity at steady state.

Trivial in our case since there are only 2 species:

- H is highest ($\approx \ln 2$) where S and R end up roughly equal, which covers most of the grid: a single dose moves the S/R split only modestly from 50/50.
- Strong inhibitor, low α (bottom right of each panel): S wins, $H \rightarrow 0$.
- Weak inhibitor with strong R protection ($\beta_{\min} = 0$, left): R leads, but H stays ≈ 0.4 – 0.6 since R never exceeds 84% of the community.
- $\beta_{\min} = 1$ (no R protection): c no longer acts on private protection, only on R's antibiotic degradation (φ); α matters most.
- Mean H at SS = 0.56.

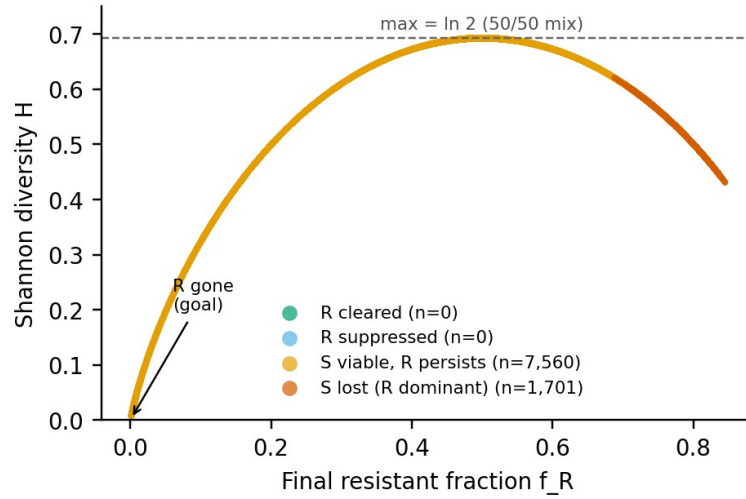


Figure 2. Diversity vs. final resistant fraction. The same low H can mean “R gone” (left) or “S gone” (right).

- With only two strains, low H is ambiguous. Always read H together with f_R .

1.2 Time to reach steady state

- SS = both populations' per-capita growth rates stay below 10^{-3} for the last 10% of the run (happens once nutrient runs out).
- All 9,261 runs reached SS within $t = 150$. Median 11.5, max 39.6.

Table 1. Time to steady state by outcome (model time units). Outcome classes use the thresholds proposed in Section 2.

Outcome	Runs	Median (IQR)	Min	Max
R cleared	0	–	–	–
R suppressed	0	–	–	–
S viable, R persists	7,560	13.0 (10.0–19.6)	7.0	39.6
S lost (R dominant)	1,701	7.5 (7.0–9.0)	5.5	16.1
All points	9,261	11.5 (9.0–17.1)	5.5	39.6

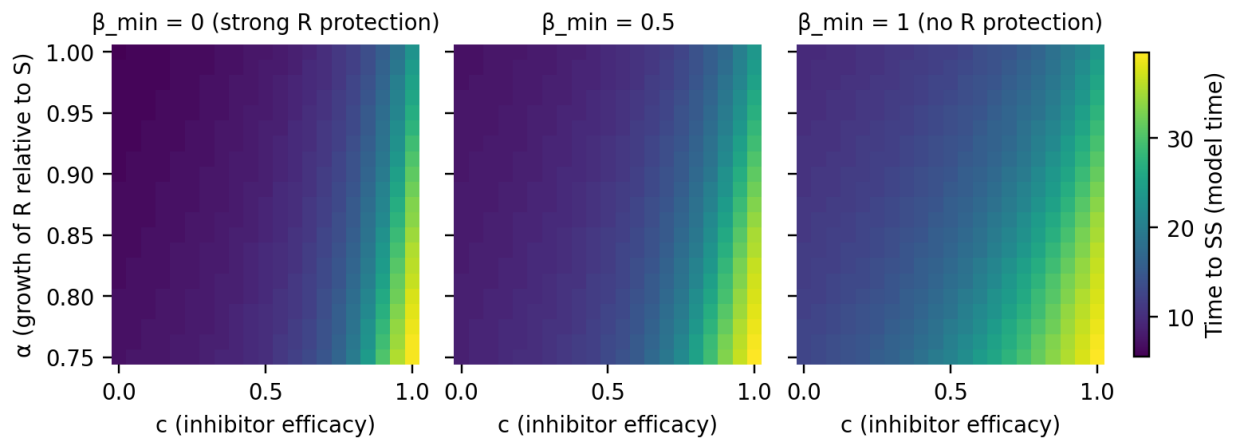


Figure 3. Time to steady state.

- R-dominant outcomes settle fastest (median ≈ 7.5): S barely grows and R uses up the nutrient quickly.
- Runs with the strongest inhibitor ($c \approx 1$) and low α settle slowest (up to 39.6): S has the deepest dip and must regrow the longest.

1.3 Final resistant fraction at steady state

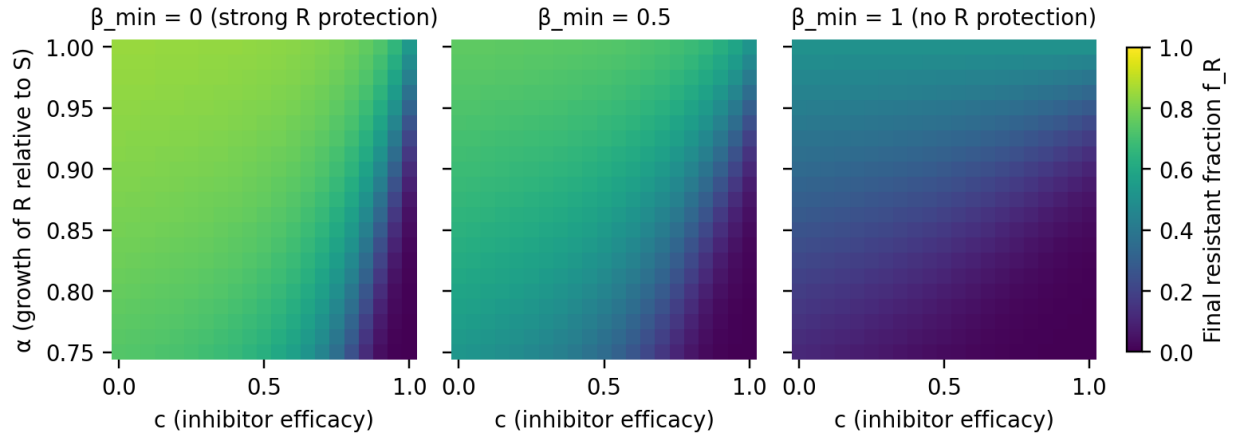


Figure 4. Final resistant fraction $f_R = n_R / (n_S + n_R)$.

- Higher c (stronger inhibitor), lower α (larger cost of resistance) and higher $\beta_{\min}$ (less R protection) all push toward S.
- The change is gradual across most of the grid; it is steepest near $c = 1$ when R's protection is strong ($\beta_{\min} = 0$).
- f_R ranges from 9.1×10^{-4} to 0.84: neither strain is ever fully removed at this dose.
- 54.0% of the grid ends with R as the majority ($f_R > 0.5$).
- A linear scale hides how small R gets in the dark region; Section 2 uses a log scale.

2. Extinction Threshold

2.1 Proposed threshold

- **R is extinct** if $f_R < 10^{-9}$ at steady state.
- **R is suppressed** (reported separately) if $10^{-9} \leq f_R < 10^{-6}$.

2.2 Rationale

- ODE populations never reach exactly zero, so a cutoff is required.
- Model density units are arbitrary, so the cutoff should be relative to the community (f_R), not absolute. The current absolute cutoff ($n_R < 10^{-4}$) equals $f_R \approx 2.5 \times 10^{-5}$.
- A saturated bacterial culture holds roughly 10^9 cells/mL, so $f_R < 10^{-9} \approx$ fewer than one R cell per mL: R is actually cleared, not just outnumbered.
- Under the current cutoff, $\sim 25,000$ R cells/mL could remain, enough to regrow once the drug is gone.
- 10^{-6} ($\sim 1,000$ cells/mL) is a useful middle tier: R is heavily outnumbered but could rebound between doses.
- Assumption: 10^9 cells/mL. Rescale both cutoffs if the experimental carrying capacity differs.

2.3 Results

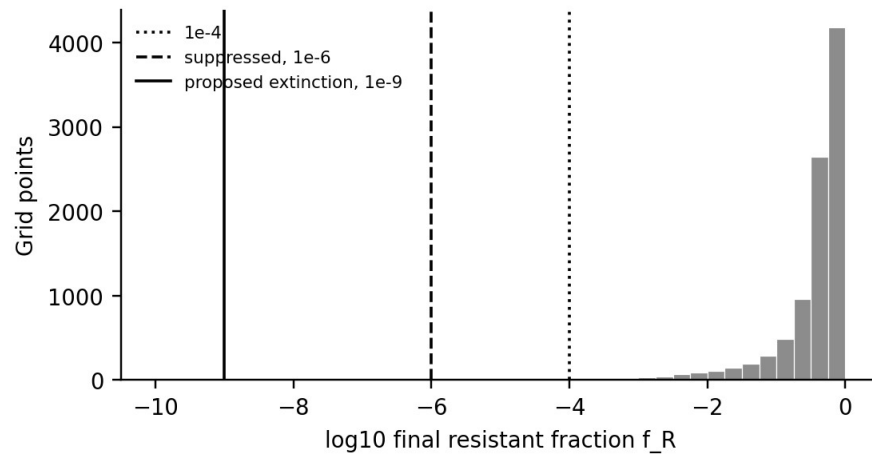


Figure 5. Distribution of final f_R (log scale) with candidate thresholds.

- f_R spans 9×10^{-4} to 0.84, far above every candidate cutoff, so no run counts as “extinct” or “suppressed” at this dose.

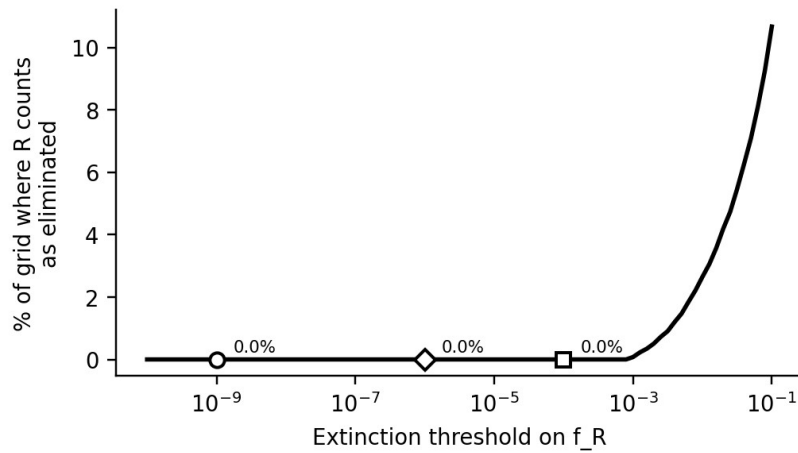


Figure 6. Share of the grid where R counts as eliminated, vs. threshold.

Table 2. Runs where R counts as eliminated, by threshold on f_R .

Threshold on f_R	Runs	% of grid
10^{-2}	245	2.6%
10^{-3}	7	0.1%
10^{-4}	0	0.0%
10^{-6} (proposed: suppressed)	0	0.0%
10^{-7}	0	0.0%
10^{-8}	0	0.0%
10^{-9} (proposed: extinct)	0	0.0%

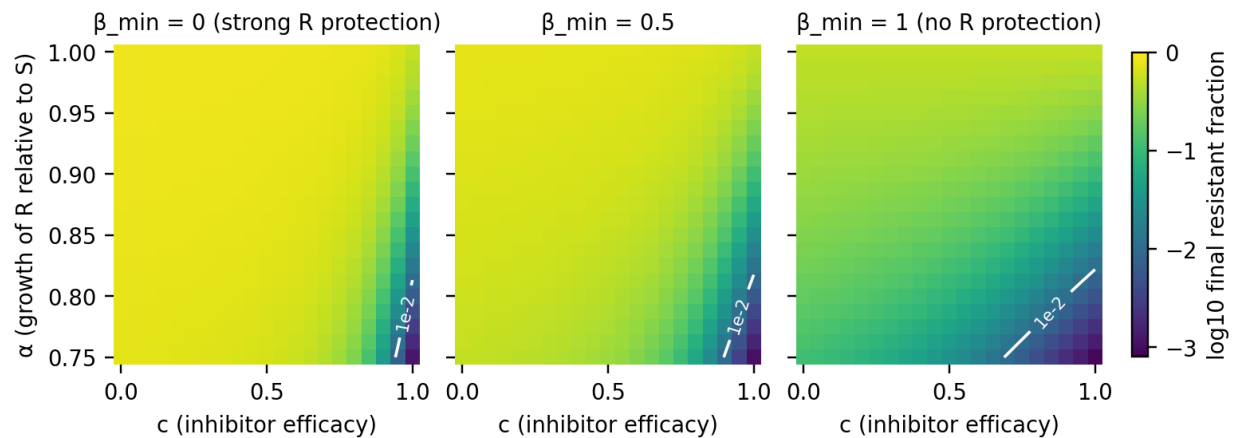


Figure 7. log10 final f_R . White contour = 10^{-2} ; neither 10^{-6} (suppressed) nor 10^{-9} (extinct) is reached anywhere on the grid.

- Proposed 10^{-9} and 10^{-6} : 0 runs. The lowest f_R (9×10^{-4}) is at $\alpha = 0.75$, $c = 1$, $\beta_{\min} = 1$.
- Only 245 runs (2.6%) reach $f_R < 10^{-2}$, all with $\alpha \leq 0.81$ and $c \geq 0.7$; 7 runs (0.1%) reach $f_R < 10^{-3}$, all at $\alpha = 0.75$, $c = 1$.
- **A single dose at $a_0 = 2.15$ does not eliminate R. This supports testing periodic (repeated) dosing or higher doses next.**

3. S Survival, Recovery and Population Loss

- **S viable:** S at SS $\geq 50\%$ of S in a drug-free run.
- **Recovery time:** first time S returns to $\geq 90\%$ of its final level after the crash.
- **Peak loss:** largest fractional drop below the starting density (S alone, and S + R).

3.1 Does S survive at a viable density?

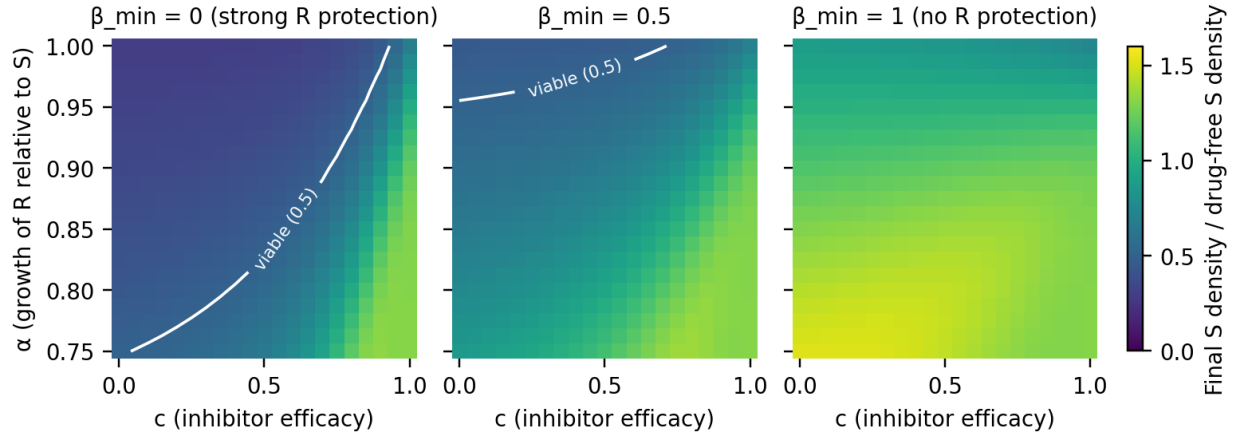


Figure 8. Final S relative to a drug-free run. White line = viability cutoff (0.5).

- S is viable in 7,560 runs (81.6%).
- S is lost only where R keeps strong protection ($\beta_{\min} \leq 0.55$) and the inhibitor is weak ($c \leq 0.9$).
- S can end at up to $\sim 1.5\times$ its drug-free level, because it takes R's share of nutrient.

3.2 Time to recovery for S

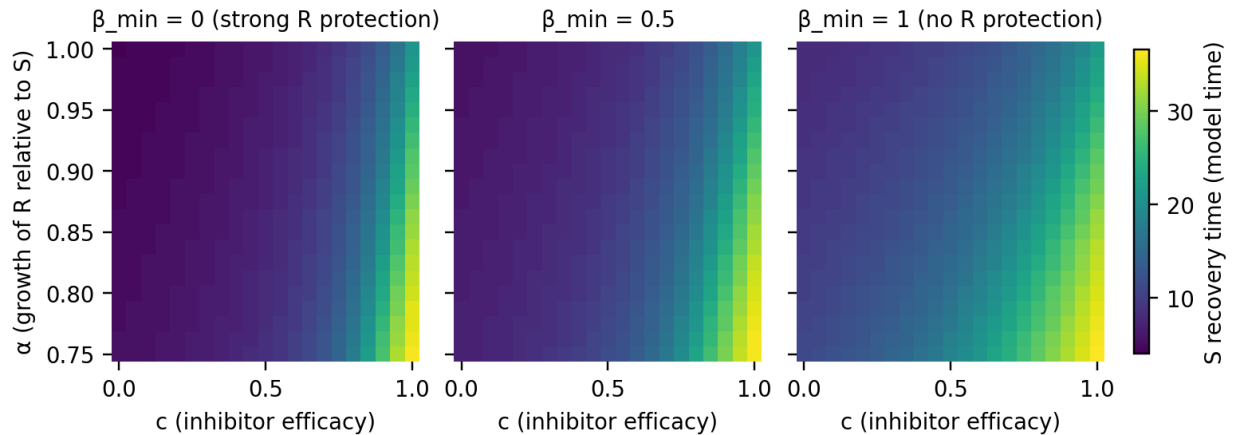


Figure 9. Time for S to recover to 90% of its final level.

- Median 10.0 (range 4.0–36.6).
- S recovers within ~ 1.5 – 3.5 time units of reaching SS in every run: S keeps growing until the nutrient runs out.
- Slowest with the strongest inhibitor and low α (up to 36.6): deeper dip, longer regrowth.
- In “S lost” runs S never returns to a viable level; its recovery time only marks where S levels off low (ignore).

3.3 Peak population loss during treatment

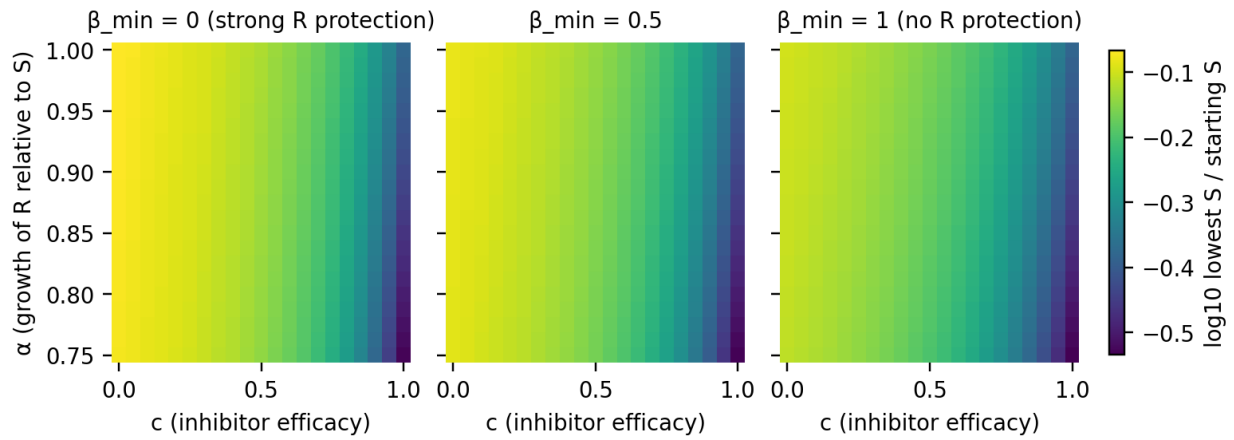


Figure 10. Lowest S reached, relative to starting S (log10). $-0.5 \approx 30\%$ of S left.

- The dip is moderate: S loses 14–71% at the bottom (median 29.9%).

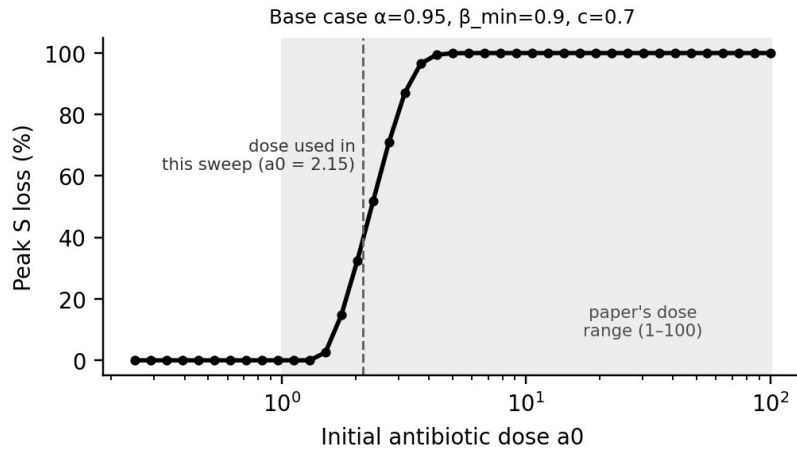


Figure 11. Peak S loss vs. initial antibiotic dose (base-case parameters).

- Dose drives the maximum loss. Loss is 0% for $a_0 \leq 1.3$, about 32% at $a_0 \approx 2$ and $\geq 99\%$ from $a_0 \approx 4.3$.
- At the paper's Fig. 2C dose ($a_0 = 2.15$) the base case loses 39% of S; a deep crash needs $a_0 \geq 4$, still inside the paper's 1–100 dose range.
- α , $\beta_{\min}$ and c only shift how deep the dip goes (from $\sim 86\%$ to $\sim 29\%$ of S left).

3.4 Diversity after S recovers

- Re-ran all 9,261 runs and recorded H at S's lowest point and at the moment S recovered.

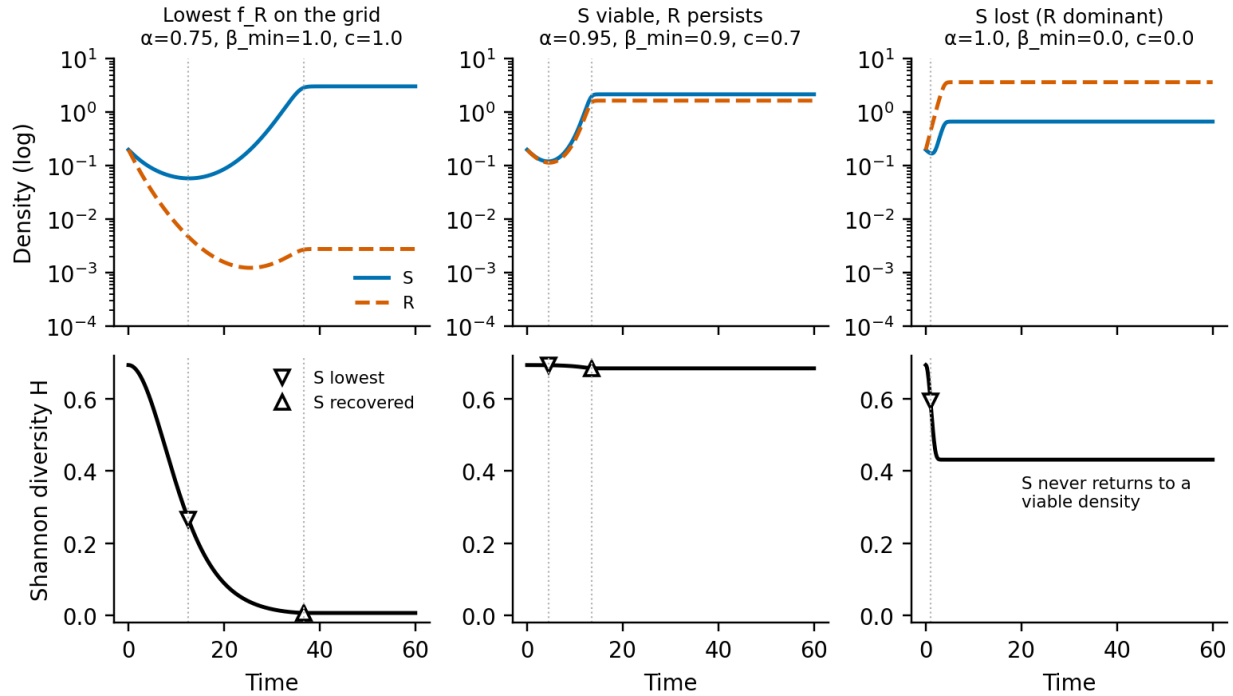


Figure 12. Three example runs. Top: S and R densities (log). Bottom: diversity, with S's lowest point (∇) and recovery ($\triangle$).

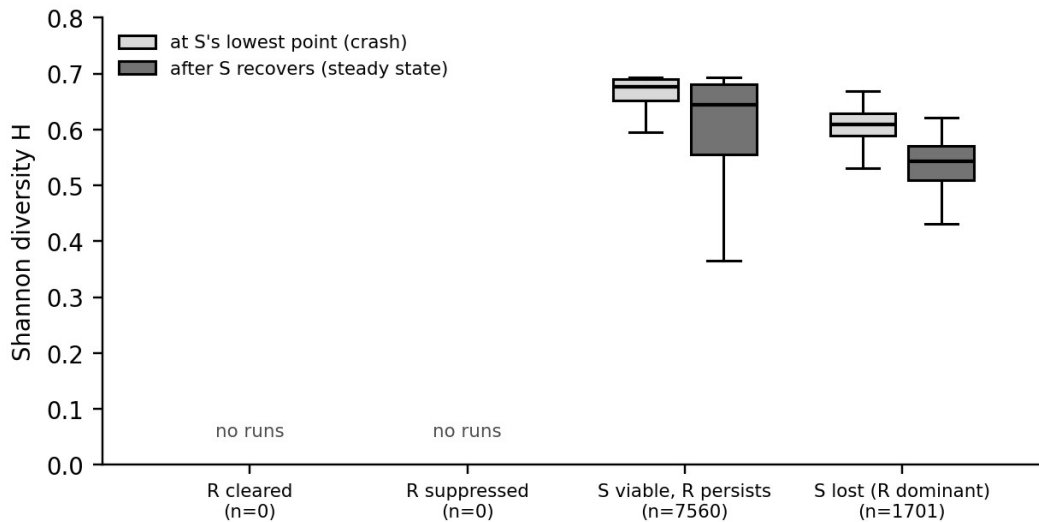


Figure 13. Diversity at S's lowest point vs. after S recovers, by outcome.

- H at recovery = H at SS (max difference 0.006): once S has recovered, the community composition is fixed.
- Diversity falls during recovery in 81% of runs: mean H 0.65 at the crash $\rightarrow$ 0.56 after recovery.
- Why: during the crash S and R are both small and similar in size; during regrowth one strain outgrows the other.
- **R cleared / suppressed:** no runs at this dose. In the lowest- f_R run (Fig. 12, left) H is still 0.27 at the bottom of the dip and falls to $\approx$ 0.01 during regrowth: R is outgrown after the crash, not wiped out during it.

- **S viable, R persists:** small median drop (0.68 → 0.65), with a long tail down to ≈ 0.01 where S pulls far ahead.
- **S lost:** H falls from 0.61 to 0.54. R keeps pulling ahead during regrowth.

Table 3. Survival, recovery, loss and diversity by outcome (medians).

Outcome	Runs	S viable	Recovery time	Peak S loss	Peak total loss	H at crash	H after recovery
R cleared	0	–	–	–	–	–	–
R suppressed	0	–	–	–	–	–	–
S viable, R persists	7,560	100%	11.0	32.89%	29.69%	0.677	0.645
S lost (R dominant)	1,701	0%	n/a*	21.12%	0.00%	0.609	0.543

* S never returns to a viable density in these runs.